

# Intro to CBS4DH

Dimitra Grigoriou

2026-04-13

136143 KU Computational Background Skills for Digital Humanities (2026S) (Group B)



## Dimitra Grigoriou

- Background in Linguistics, English Lit.
- Since August 2024 at IHB
- Data modeling and Technical Support
- Currently:
  - TEI-C Board Member
  - sources on Habsburg-Ottoman diplomacy  
QHOD
  - Minutes of the Councils of Ministers

## Index

- How this course works
- Get to know each other
- CBS4DH: what does that mean?
- Three values in DH
  - Collaboration
  - Openness
  - Versatility

## How this course works

- Six parts
  - File formats and encodings (13.04.)
  - Regular expressions (15.04.)
  - Minimal introduction to web technologies (22.04.)
  - Command line (23.04.)
  - Git (29.04.)
  - Tools and resources for Digital Humanities (30.04.)
- Course material: [https://github.com/acdh-oeaw/Teaching\\_CBS4DH/tree/2026S](https://github.com/acdh-oeaw/Teaching_CBS4DH/tree/2026S)



## How this course works

- **Attendance** is extremely important! **Absence:** 1,5 Modules
- Complete the **exercises** on **Moodle**. **Deadline:** Monday May 4<sup>th</sup> 2026
- Please **follow along** with the tutorials
- **Ask questions**
- **Check** if you have installed everything
  - Visual Studio Code: <https://code.visualstudio.com/>
  - (only Windows users) Gitbash: <https://gitforwindows.org/>
  - Create a GitHub account: <https://github.com/join>

## Two questions for you

- What are you studying / have you studied previously?
- What operating system do you use?

# CBS4DH

Computational Background Skills  
for  
Digital Humanities

# CBS4DH

Computational Background Skills  
for  
**Digital Humanities**

## What is Digital Humanities?

- The application of digital methods to research in the Humanities
- Varying degree of use of digital methods
- <https://whatisdigitalhumanities.com>

**What Is Digital Humanities?**

New quote

- Around 800 definitions of DH
- [Raw data](#)

# What Is Digital Humanities?

New quote

The best way I've found to describe what Digital Humanities is, in first place to myself, is using a quote from Thomas Pynchon's Crying of Lot 49: 'You know what a miracle is ... another world's intrusion into this one' **Federico Meschini, De Montfort University**

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**NB:** Click the New quote button to display a new random quote. Quotes were pulled from participants from the Day of DH between 2009-2014. As of January 2015, the database contains 817 rows and randomly selects a quote each time the page is loaded. If you want to do something cool with the data, I am providing the code and data here.

Made by Jason Heppler. Problems? Questions? I'm @jaheppler on Twitter.

## Not only Humanities

- Social Sciences and Humanities (**SSH**)
  - For example, SSHOC ([Social Sciences and Humanities Open Cloud/Cluster](#))
- Political science
- Business
- Broad application of certain techniques/research areas
  - e.g. Natural Language Processing
- And many others

# CBS4DH

Computational Background Skills  
for  
Digital Humanities

# CBS4DH

Computational Background Skills  
for  
Digital Humanities

„Push your changes to the repo“

CSS

```
^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$
```

HTML

„You can write a shell script to do this“

131.130.70.8

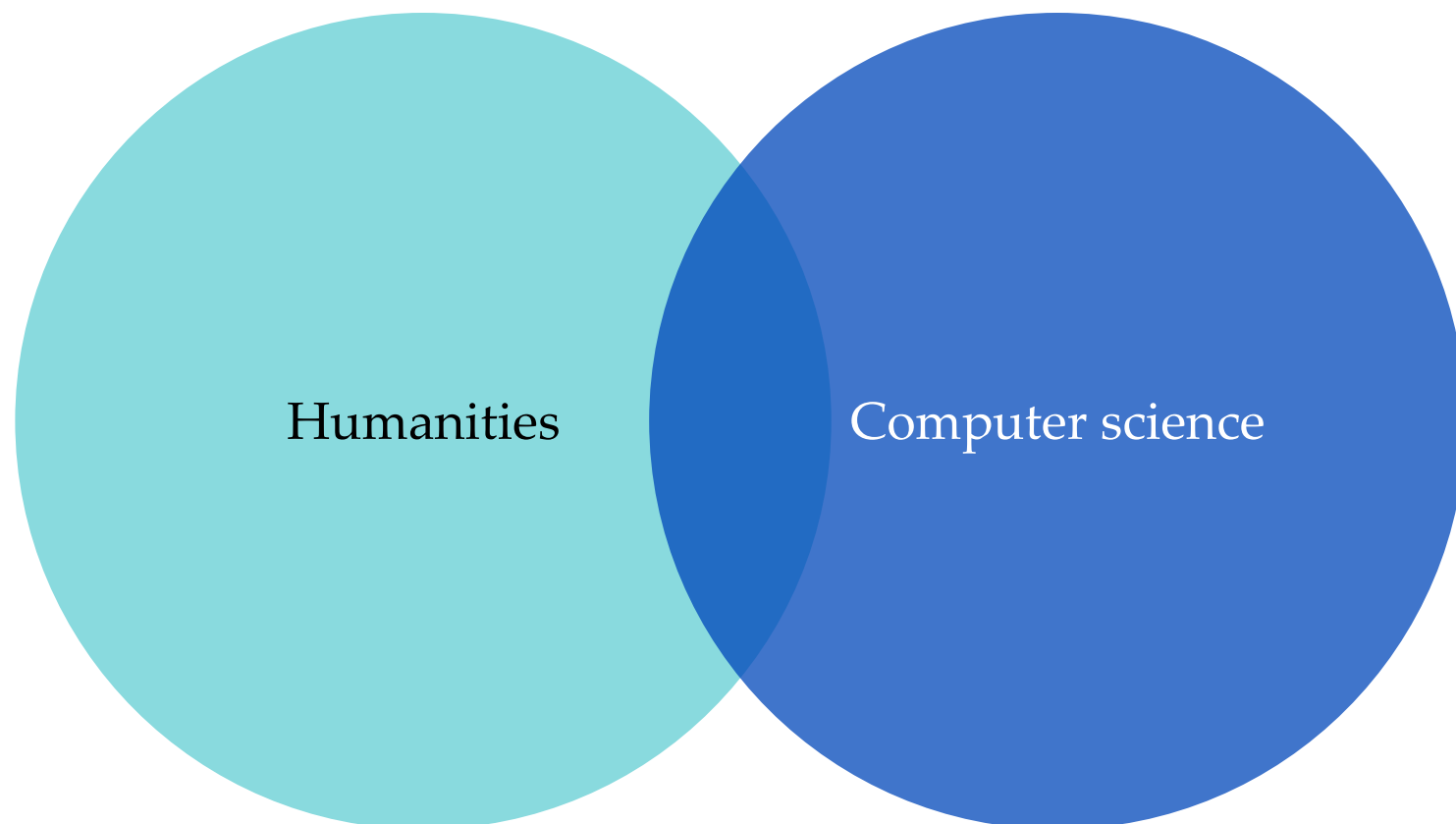
```
Alice,25,New York,Engineer
Bob,30,Los Angeles,Doctor
Charlie,35,Chicago,Artist
```

# CBS4DH

Computational Background Skills  
for  
Digital Humanities

# CBS4DH

Computational Background Skills  
**for**  
Digital Humanities



## Some things you can do with these skills

- Print a list of all files in a directory
- Find all instances of “colour” and “color” with a single search
- Keep a change history of your document
- Create a basic website
- Find new opportunities to learn
- ... and many more!

# Three ‚values‘ in DH

# Collaboration

## Communities and collaboration

- DH project = no one-man show!
- Collaboration between different actors, stakeholders and communities
  - Digital Humanists
  - Developers
  - More “traditional” scholars
  - Different departments / institutions
  - Broader audience



One Man Band, CC BY 2.0, Created by [Iarekt](#)

## Digital Humanities Centers / Institutes in Austria

- [Digital Humanities @ University of Vienna](#)
- [Austrian Centre for Digital Humanities](#) (at ÖAW)
- University of Graz – [Institut für digitale Geisteswissenschaften](#)
- Lists
  - [EADH Digital Humanities Centers](#)
  - [NYU List of Digital Humanities Centers](#)

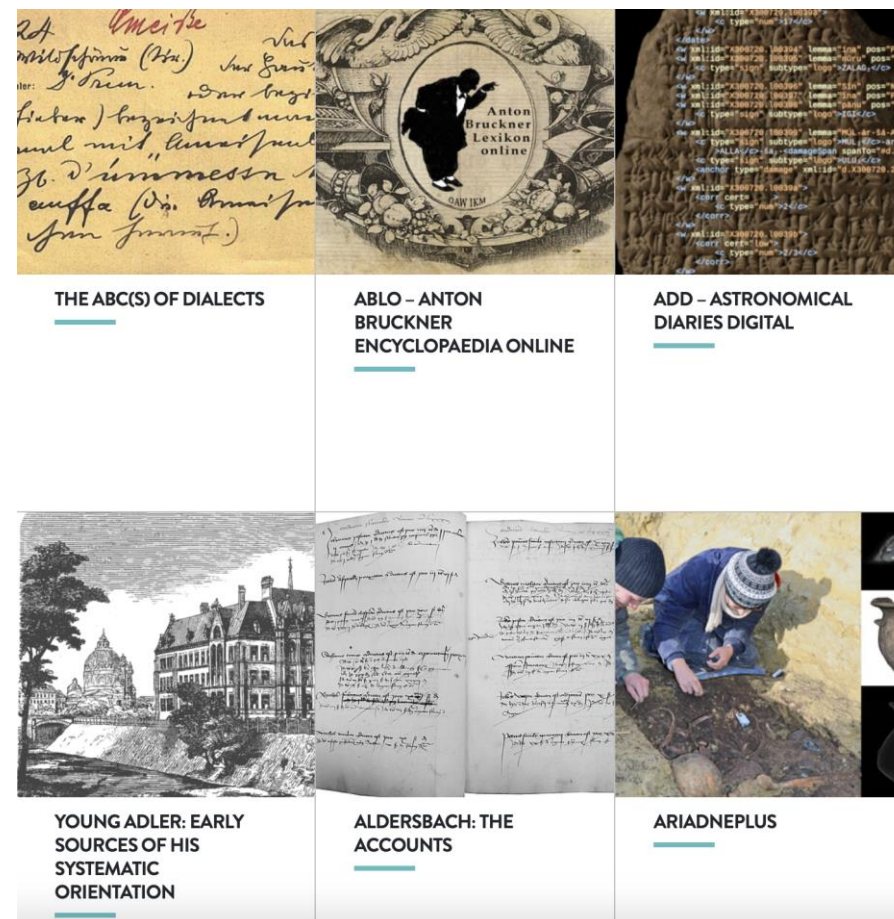
## Austrian Centre for Digital Humanities (ACDH)

- Founded in 2015, concept from 2013
- Mission: *fostering the use of digital methods and tools in the humanities research in Austria*
- Rapid growth
  - From 12 in 2015 to 141 in 2024
  - ~95 [projects](#)



# Austrian Centre for Digital Humanities (ACDH)

- Four research units
  - Linguistics
  - Literature and Text
  - Musicology
  - DHRI – Digital Humanities Research & Infrastructure
- Both a research *and* a service-providing institution  
(What we offer)



## DH associations

- DHd – Digital Humanities im deutschsprachigen Raum  
<https://digitalhumanities.de>
- EADH – European Association for Digital Humanities  
<http://eadh.org>
- ADHO – Alliance of Digital Humanities Organizations  
<https://adho.org> (list of partners: <https://adho.org/membership-individual/>)
- MLA2S – Machine Learning at the Austrian Academy of Sciences  
<http://www.oeaw.ac.at/mla2s>

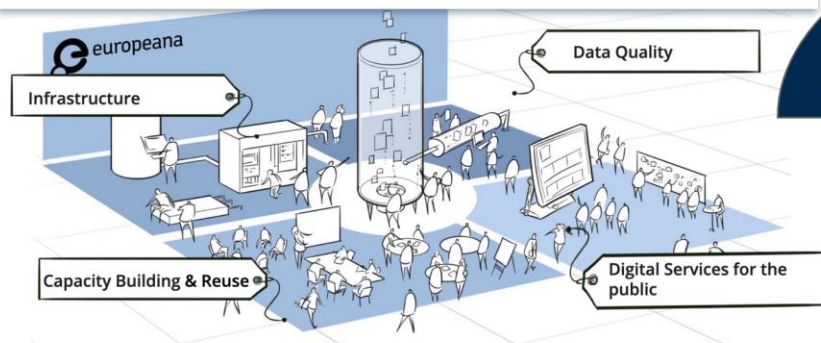


## Research Infrastructures

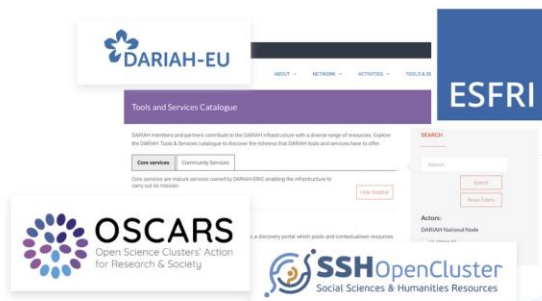
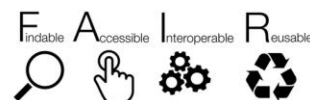
- “facilities that provide resources and services for research communities to conduct research and foster innovation” ([European Commission](#))
  - No humans mentioned!
- Networks of people and expertise
- Both bottom-up and top-down
  - Exchange of ideas and knowledge sharing
  - Consolidation, harmonization (standards, best practices)
  - Provision of services

# The European Research Landscape

**Role: Interoperable Data Aggregation  
(beyond metadata)**



**Role: State of the Art processing of  
Cultural Heritage Data (e.g AI-based)**



Social Sciences & Humanities Open Marketplace

Discover new and contextualised resources for your research in Social Sciences and Humanities: tools, services, training materials, workflows and datasets. Read more...

All categories Search

**Role: Publishing of tools; methods  
and derived datasets for reuse**



## DARIAH

- European *Digital Research Infrastructure for the Arts and Humanities*
- Strong voice for the DH community on the European level
- Lines of action
  - Common policies & guidelines
  - Dissemination and discoverability of resources ([SSH Open Marketplace](#))
  - Knowledge sharing ([DARIAH Campus](#))
  - Integration with other initiatives, e.g. European Open Science Cloud ([EOSC](#))

# Social Sciences & Humanities Open Marketplace

Discover new and contextualised resources for your research in Social Sciences and Humanities: tools, services, training materials, workflows and datasets. [Read more...](#)

All categories



Search

[SSH Open  
Marketplace](#)



Search

[Home](#) / [Tools & services](#) / [Gephi](#)

# Gephi

[Copy to clipboard](#)

Gephi is the leading visualization and exploration software for all kinds of graphs and networks.

Gephi is open-source and free. Runs on Windows, Mac OS X and Linux.

[Go to Tool or service](#)

## Details

### ACCESS

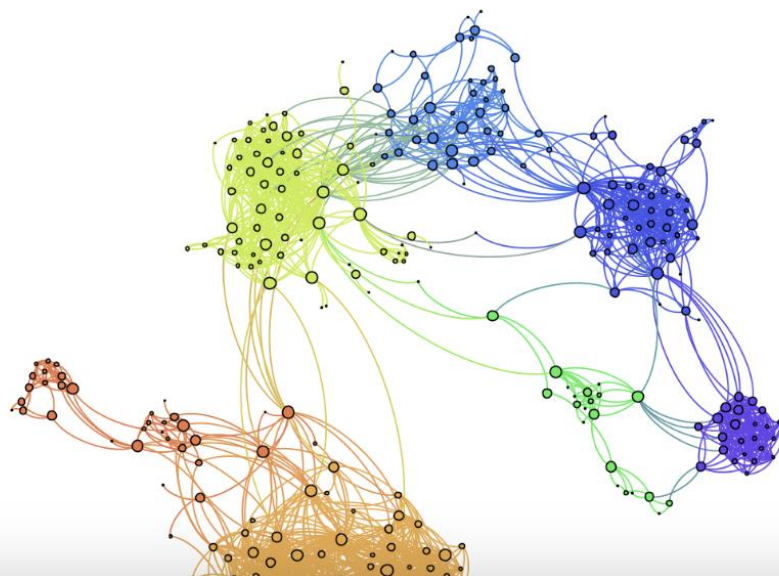
License [Common Development and Distribution License 1.0](#) [GNU General Public License v3.0 only](#)

Terms Of Use [Free](#)

### CATEGORISATION

Activity [Capturing](#) [Modeling](#) [Network Analysis](#) [Analyzing](#) [Visual Analysis](#) [Data Visualization](#) [Relational Analysis](#) [Spatial Analysis](#) [Content Analysis](#) [Discovering](#)

Keyword [Models](#) [graphs](#) [french](#) [R](#) [faceted browser](#) [graph streaming](#) [recommended](#)



## CLARIN

**CLARIN**  
Common Language Resources and  
Technology Infrastructure



- *Common Language Resources and Technology Infrastructure*
- Easy and sustainable access to a broad range of language data and tools
- Distributed research infrastructure, with [centers all over Europe](#)
- Services
  - [Content search](#) (search for specific patterns across collections)
  - [Virtual Language Observatory](#) (explore language resources and tools)
  - [Language Resource Switchboard](#) (process your data with specialized tools)
  - [Learning Hub](#) and [DH Course Registry](#) (find learning opportunities in DH)

# DIGITAL HUMANITIES COURSE REGISTRY



INFO



FILTER

SORT

Find a course directly by name or university

| NAME                                                            | UNIVERSITY                      | PLACE                    | DATE                      | TYPE               |
|-----------------------------------------------------------------|---------------------------------|--------------------------|---------------------------|--------------------|
| MA in Digital Humanities                                        | King's College London           | London, United Kingdom   | 24 Sept ↻<br>1 years      | Master Programme   |
| MSc in Digital Scholarship                                      | University of Oxford            | Oxford, United Kingdom   | 9 Oct ↻<br>11 months      | Master Programme   |
| Digital Stylometry                                              | Freie Universität Berlin        | Berlin, Germany          | 17 Apr 2025<br>1 semester | Course             |
| Introduction to the Digital Humanities                          | Freie Universität Berlin        | Berlin, Germany          | 14 Apr ↻<br>1 semester    | Course             |
| Postgraduate Certificate in Digital Humanities and Culture      | Trinity College Dublin          | Dublin, Ireland          | 5 Sept 2025<br>2 semester | Course             |
| Postgraduate Diploma in Digital Humanities & Culture            | Trinity College Dublin          | Dublin, Ireland          | 5 Sept ↻<br>2 semester    | Course             |
| MPhil in Digital Humanities and Culture                         | Trinity College Dublin          | Dublin, Ireland          | 5 Sept ↻<br>1 years       | Master Programme   |
| Swedish National Research School in Digital Humanities: Dat ... | Uppsala University              | Uppsala, Sweden          | 25 Sept ↻<br>12 months    | PhD Programme      |
| Digital Humanities                                              | Higher School of Economics Perm | Perm, Russian Federation | 1 Sept ↻<br>2 years       | Master Programme   |
| Language and computers                                          | Universitetet i Bergen          | Bergen, Norway           | 15 Jan ↻<br>1 semester    | Bachelor Programme |
| Master in Textual Heritage & Digital Humanities [Máster en ...  | Universidad de Salamanca        | Salamanca, Spain         | 23 Sept ↻<br>2 semester   | Master Programme   |
| Corpus-Driven Lexicology                                        | Mykolas Romeris University      | Vilnius, Lithuania       | 1 Feb ↻<br>5 months       | Bachelor Programme |



Imprint

FAQ

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## DH Course Registry



## Austrian DH landscape

- CLARIAH-AT – [clariah.at](http://clariah.at)
  - CLARIN + DARIAH at the Austrian level
- DiTAH – [ditah.at](http://ditah.at)
  - *Digitale Transformation der österreichischen Geisteswissenschaften*
  - Four-year project (2020-2024) for the establishment of Digital Humanities in Austria
- Funding schemes (e.g. CLARIAH-AT [Project funding](#), [Funding for Early Career Researchers](#), [Transkribus Scholarship](#))
- Next DHd 2027 Conference will take place in Marburg!  
(1<sup>st</sup> – 5<sup>th</sup> March 2027)



# CLARIAH-AT Partners



## Cultural Heritage (CH) Infrastructures

- Improve the availability of CH objects for research and beyond
- Particularly relevant for [GLAM](#) sector (Galleries, Libraries, Archives and Museums)
- Austrian level: [Kulturpool](#)
- European level: [Europeana](#)
- ECCCH  
([European Collaborative Cloud for Cultural Heritage](#))



ALBERTINA

ARCHE

nhm  
naturhistorisches  
museum wien

österreichische **mediathek**  
audiovisuelles archiv · technisches museum wien

Kultur  
pool

Austrian CH aggregator



europaana

European CH aggregator



Slovakiana



Swedish Open Cultural  
Heritage



CulturaItalia



National Heritage Institute,  
Bucharest



Greek Aggregator  
SearchCulture.gr



Gallica



... and many others

[Full list of Europeana aggregators](#)

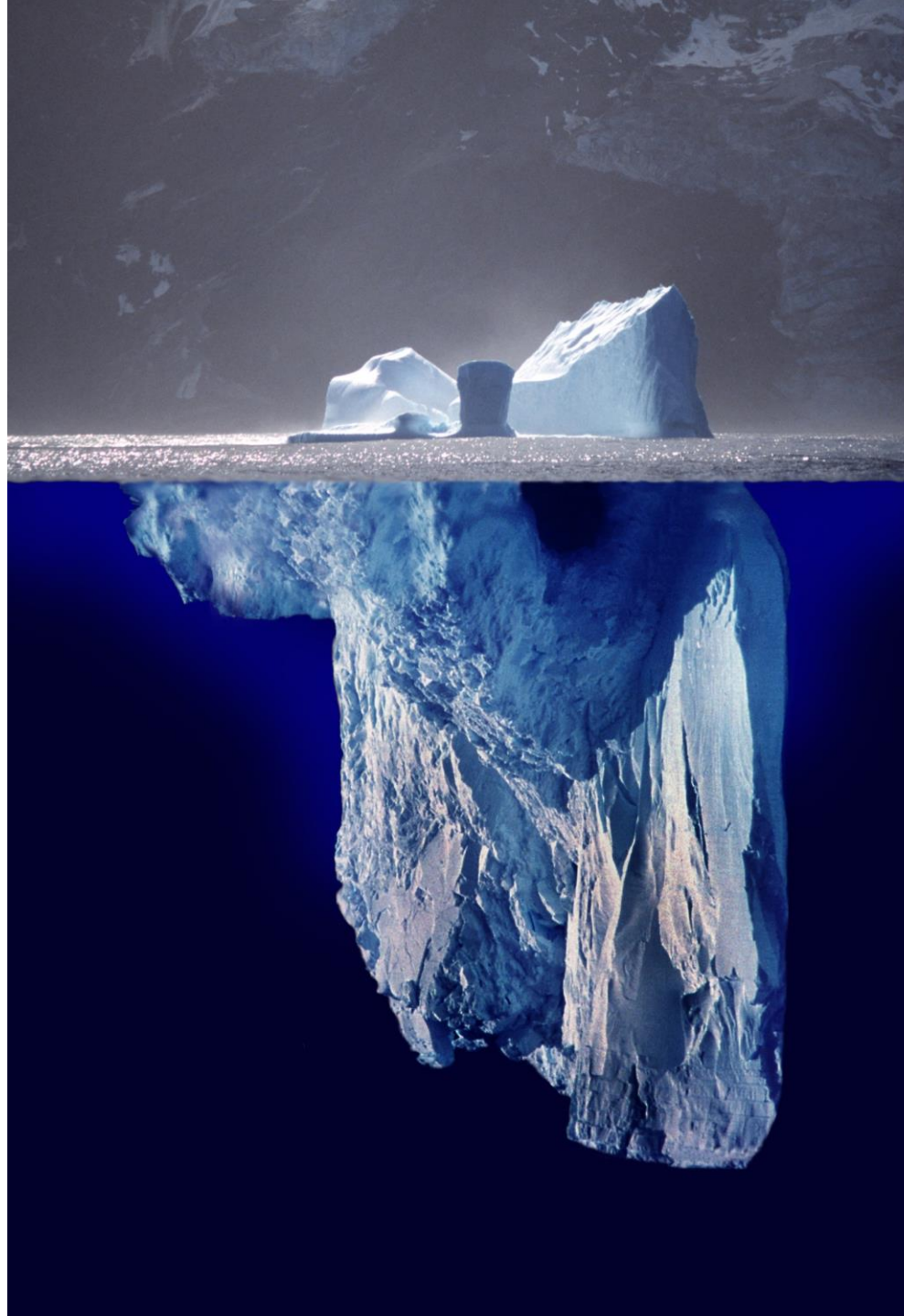
# Openness

Website  
Book/Article  
Digital edition

**Below the surface...**

**Data**

Scans  
Photos  
Database  
3D models  
Transcriptions  
Audio recordings  
Video recordings  
Reports and notes  
...



CC BY-SA 3.0, created by Uwe Kils

## Data is/are important

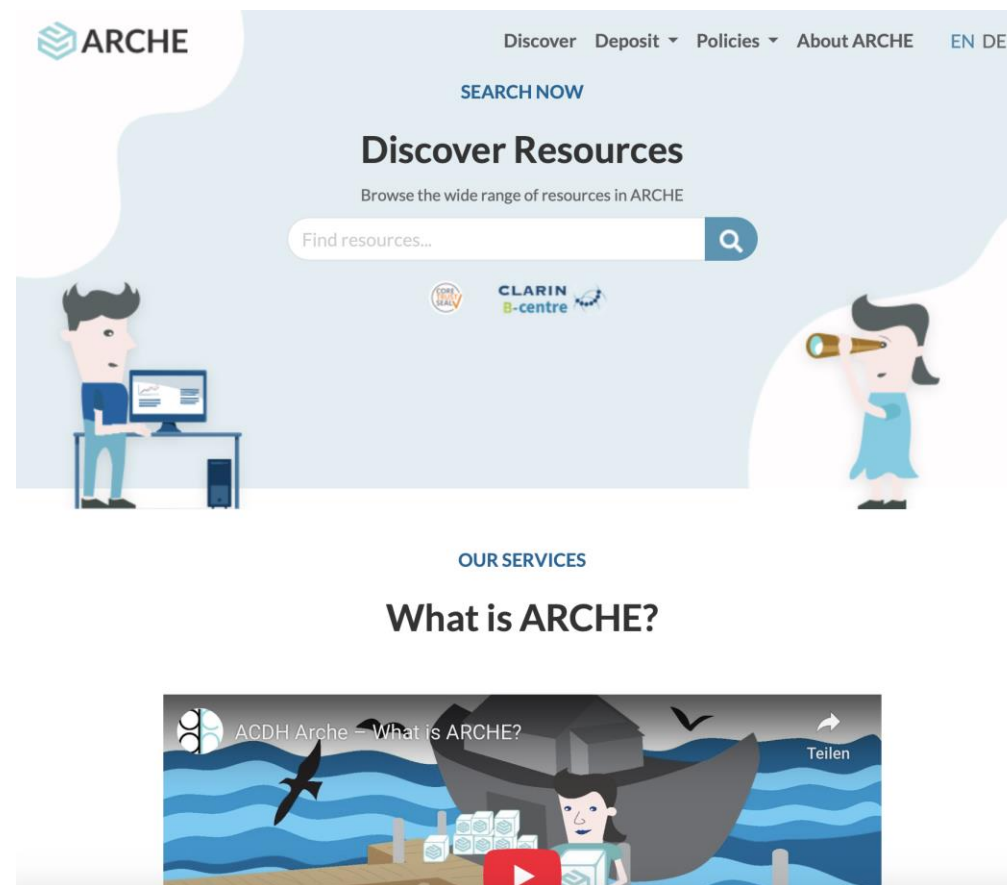
- Data is the backbone of your research, even in the Humanities
- Open data allows you to validate your results
- Open data allows others to build on your research
- **Data first!** Research data as one of the most important outputs of projects
  - Datasets as first-class citizens in the research landscape
  - Can be published in repositories like [Zenodo](#) ...
  - ... and referenced as publications (e.g. with a [DOI](#))

zenodo



## ARCHE

- Repository for digital research data in the Humanities
- Developed and hosted by ACDH
- Curation team that helps in deposition process, checks & dissemination
- Integration with CLARIN VLO and other European catalogues



# Open Data

- FAIR Principles
  - Findability
  - Accessibility
  - Interoperability
  - Reusability
- Governments and local authorities
  - <https://www.data.gov.uk>
  - <https://www.data.gv.at>

data.gv.at

56072 Datensätze  
770 Anwendungen  
1356 Organisationen

de en

Umfrage zur Verbesserung von data.gv.at gestartet

Daten > Anwendungen > Infos > News >

Suche Suchbegriff ... erweiterte Suche

Themen

- Bevölkerung und Gesellschaft
- Bildung, Kultur und Sport
- Energie
- Gesundheit
- Internationale Themen
- Justiz, Rechtssystem und öffentliche Sicherheit
- Landwirtschaft, Fischerei, Forstwirtschaft und Nahrungsmittel
- Regierung und öffentlicher Sektor
- Regionen und Städte
- Verkehr
- Umwelt
- Wirtschaft und Finanzen



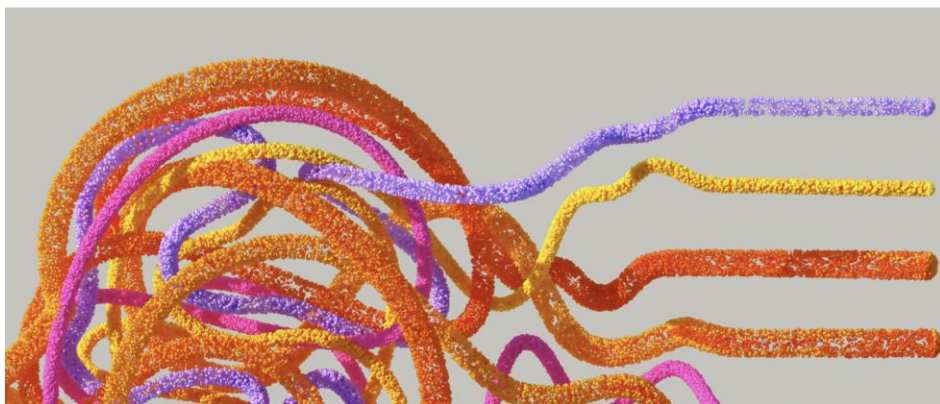
## “Openness” in modern trends


[Join us](#)
[Sign in](#)

EMERGING TECHNOLOGIES

### What is open-source AI and how could DeepSeek change the industry?

Feb 5, 2025



Bundesministerium  
Bildung, Wissenschaft  
und Forschung



Home > ... > Strategien > Open Innovation Strategie

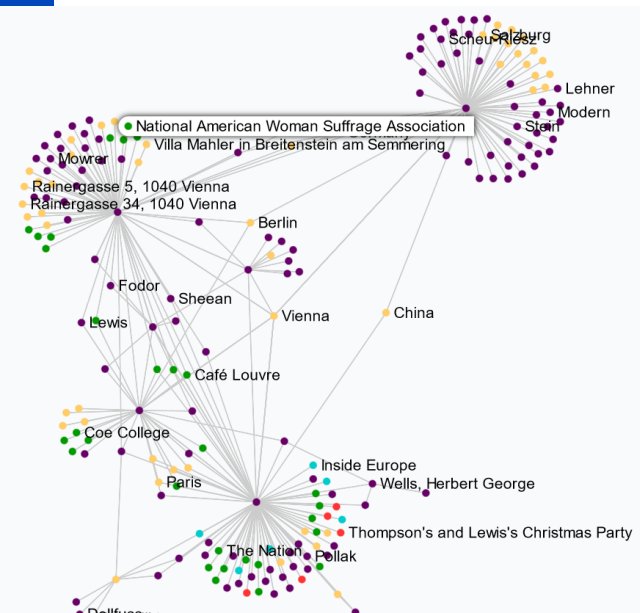
### Open Innovation Strategie

Österreich ist der erste europäische Mitgliedsstaat, der eine umfassende nationale Open Innovation Strategie entwickelt hat.

Im Zuge eines breiten Beteiligungsprozesses, in welchem nicht nur Stakeholder aus Forschung, Wissenschaft und Wirtschaft sondern auch die Zivilbevölkerung eingebunden waren, wurde im Jahr 2016 eine Open Innovation Strategie für Österreich erstellt. Diese umfasst neben einer Analyse des nationalen und internationalen Status Quo zum Thema Open Innovation auch eine Vision für 2025 und konkrete Maßnahmen zur Umsetzung.



# Versatility



in sehr guter Maske.  
14/11 Vm. bei Gisa. – Telegr. an Hauptmann,  
– Ins Grand Hotel zu Dr. Sigmund Zeisler, seit Jahrzehnten  
nicht gesehen; seine Frau Fanny Bloomfield-Zeisler, Paul Zeisler und  
seiner Sohn. Erzählt mir von seinen Krankheiten und Operationen. Über Antisemitismus u. dergl., amerik. Verhältnisse. –  
Forest in der Tram; der sehr entzückt von Heini sprach. –  
In der N. W. Bühne, wegen Sitzen zu W. L.; – spreche fl

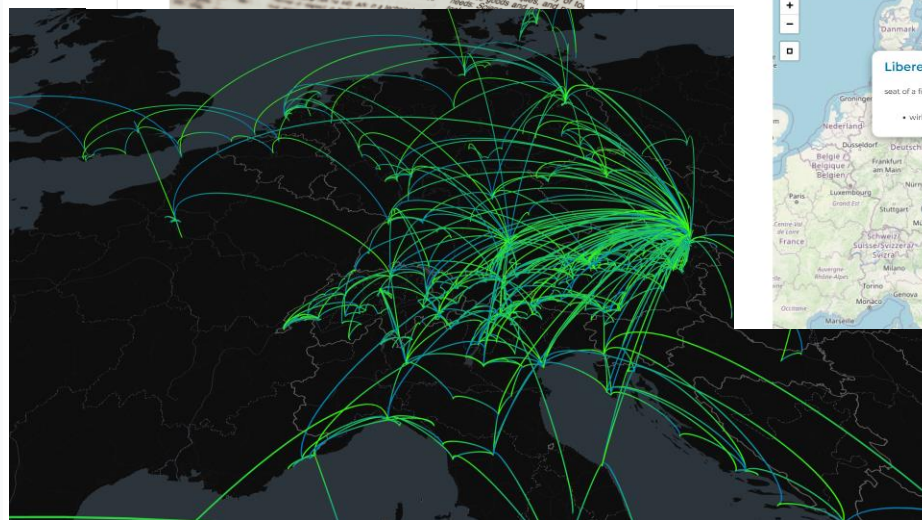
## Workflows



## Create a dictionary in TEI

## Workflow steps (5)

- 1 Build the model of the dictionary
- 2 Create a corpus of useful resources for the dictionary.
- 3 Setting up the editing environment
- 4 Setting up the publishing environment
- 5 Provide for long-term availability



## 3 Setting up the editing environment

Choose a tool for editing the dictionary, and also consider the back-end. For the latter one, we can discern two prototypical approaches: systems based on relational databases (MySQL, PostgreSQL etc.) and native XML databases (eXist, BaseX).

## Related items (4)



## Dictionary Writing Systems \&amp; Corpus Query Systems

No description provided.

[Read more](#)



## Introducing Lexonomy: an open-source dictionary writing and publishing system

This demo introduces Lexonomy  
www.lexonomy.eu, a free, open-source, web-based  
dictionary writing and publishing system. In...

[Read more](#)



## BaseX Documentation

BaseX is a light-weight, high-performance and  
scalable XML Database and an XQuery 3.1  
Processor with full support for the W3C Update...

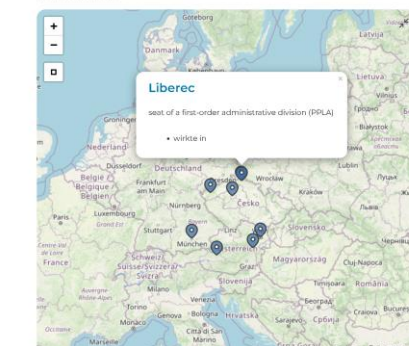
[Read more](#)



## OBL

## Streit, Andreas

Streit Andreas, Architekt. Geb. Habendorf, Böhmen (Stráž nad Nisou, Tschechien), 15. 7. 1840; gest. Reichenau (Reichenau an der Rax, NO), 20. 1. 1916; röm.-kath.



Zur Edition Kalender Einträge Personen Werke Orte



Dienstag, 14. November 1922

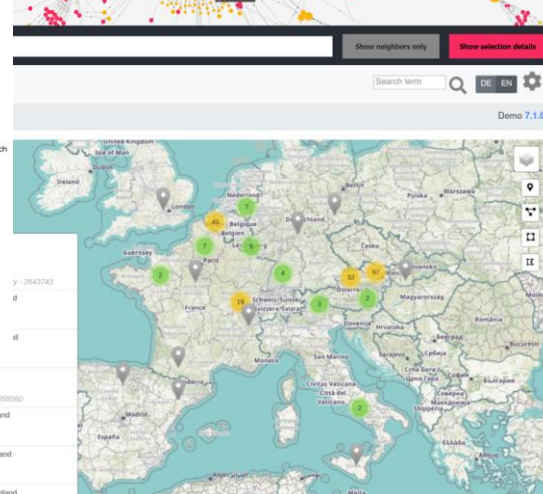
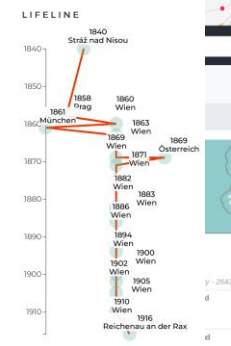
14/11 Vm. bei Gisa. — Telegr. an Hauptmann zu seinem 60. —

— Ins Grand Hotel zu Dr. Sigmund Zeisler, seit Jahrzehnten nicht gesehen; seine Frau Fanny Bloomfield-Zeisler, Paul Zeisler und seinen Sohn. Erzählt mir von seinen Krankheiten und Operationen. Über Antisemitismus u. dergl., amerik. Verhältnisse. —

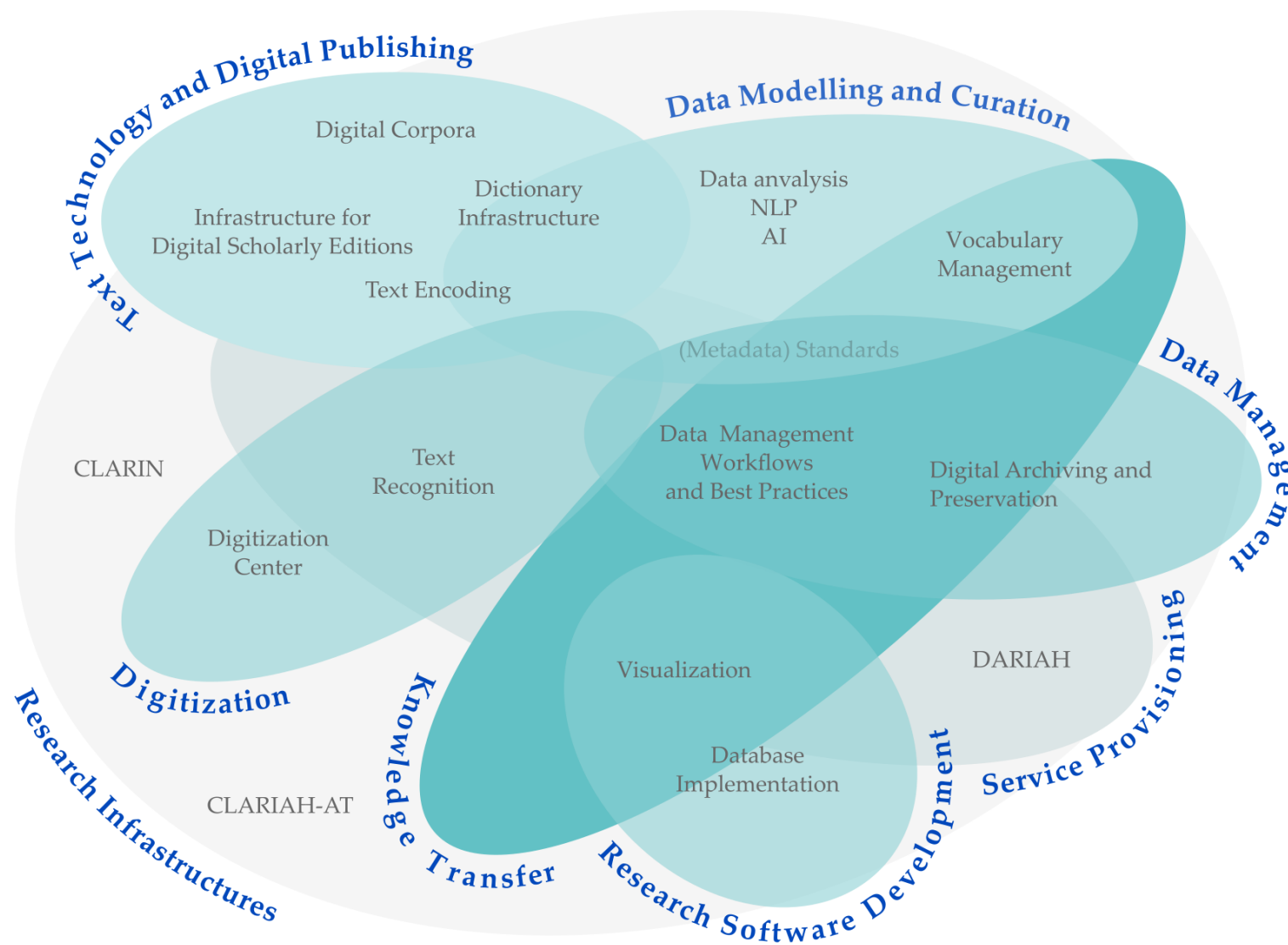
Forest in der Tram; der sehr entzückt von Heini sprach. —

In der N. W. Bühne, wegen Sitzen zu W. L.; — spreche flüchtig Korff. —

Mit Kolap Rmdth. „Weber“. Heini spielte den Kandidaten und den zweiten jungen Weber. Gute Imitation von Martin. Das Stück wirkte wieder außerordentlich. Ein Gestalter



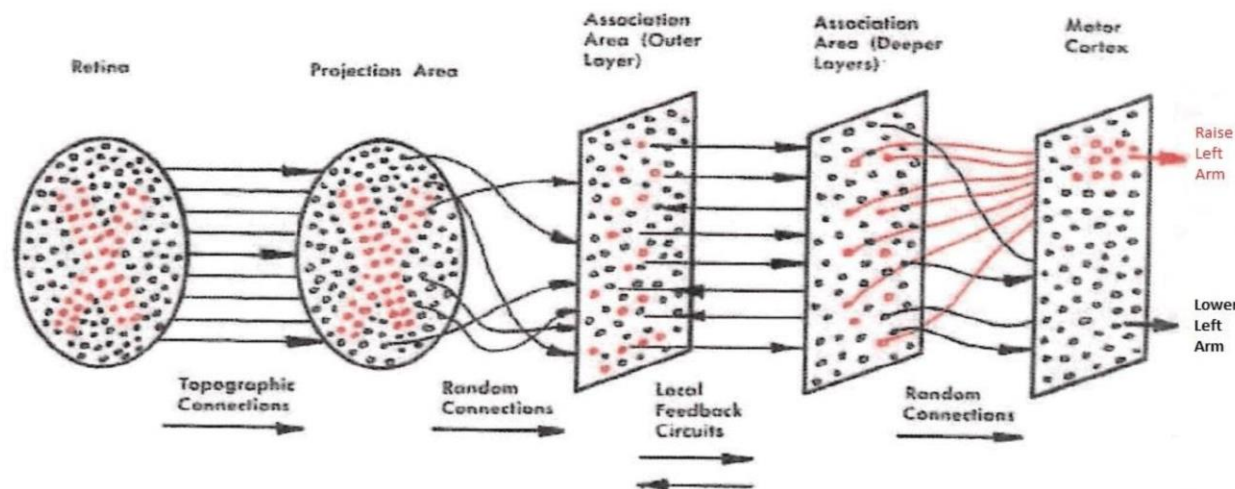
# Areas of Expertise



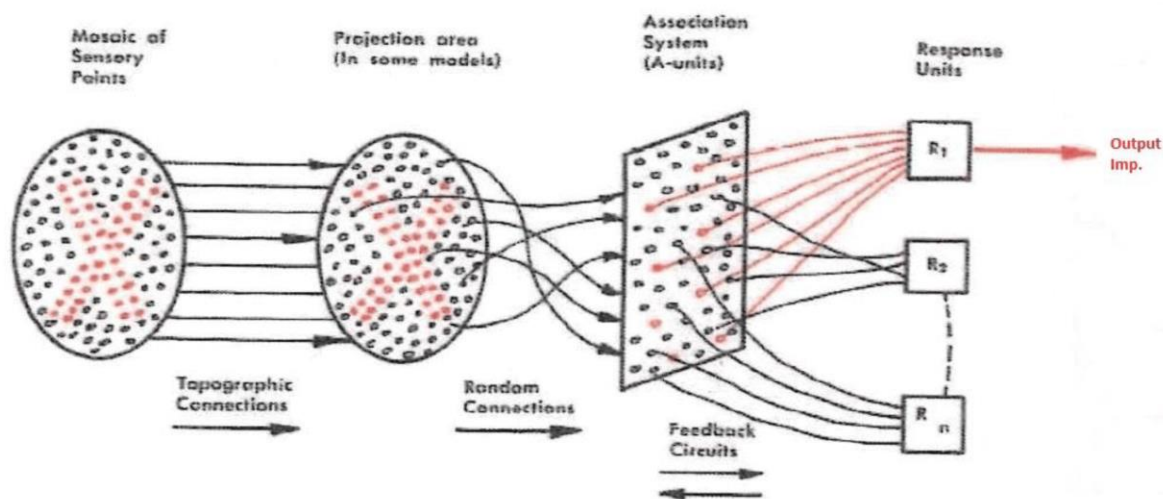
# Technology trends

## Perceptron

Rosenblatt, F. *The Design of an Intelligent Automaton*, Research Reviews, Office of Naval Research. Washington, October 1958, 7.



**FIG. 1** — Organization of a biological brain. (Red areas indicate active cells, responding to the letter X.)

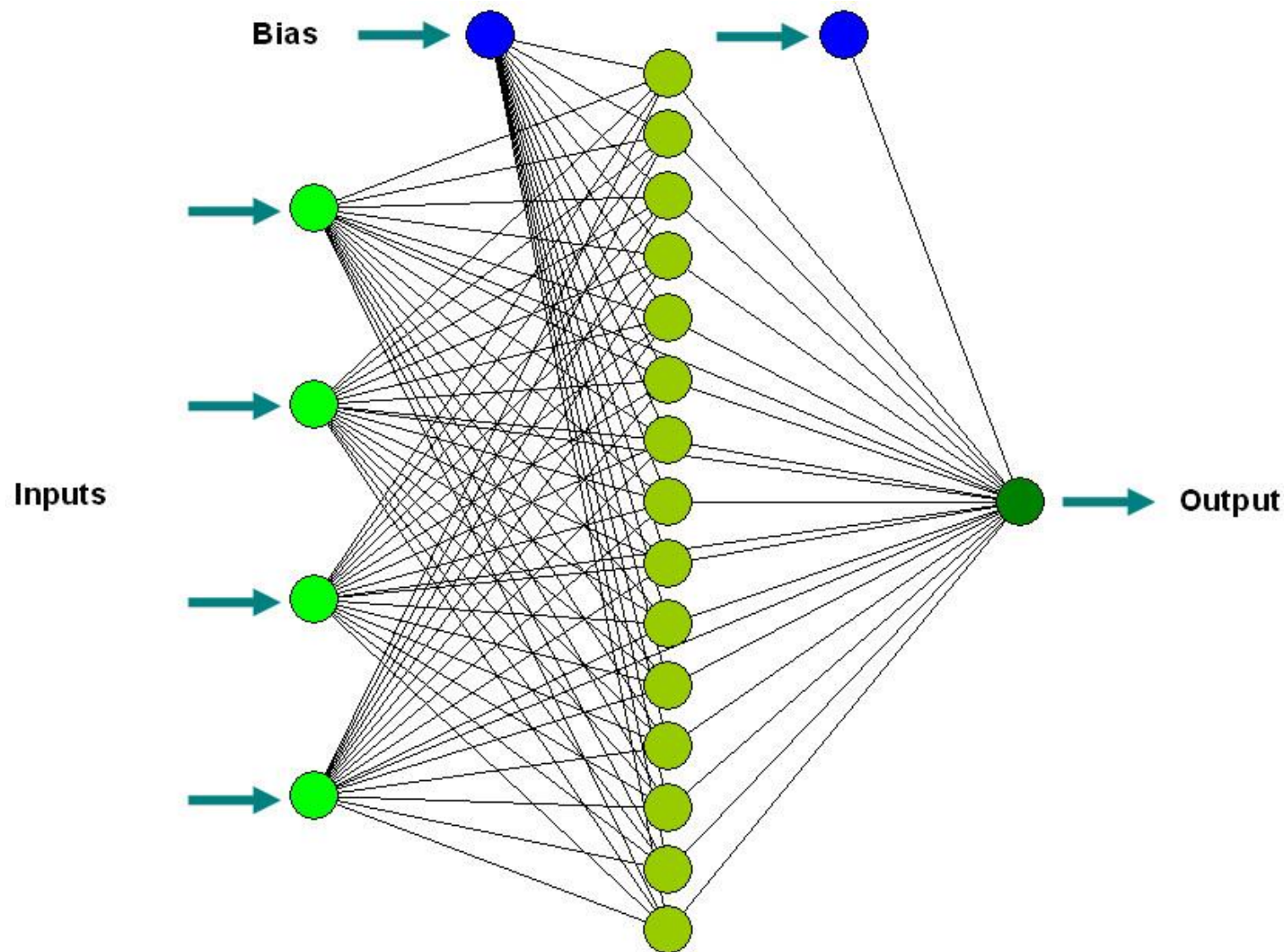


**FIG. 2** — Organization of a perceptron.

## Technology trends

### Deep neural network

Basis for modern  
Artificial  
Intelligence



## Technology trends



## The Semantic Web

*Scientific American,*  
May 2001  
Volume 284, Issue 5

## Technology trends

- Omnipresence of the Internet
- Mobile devices
- Virtual / Augmented Reality
- Widespread use of Artificial Intelligence  
(machine learning, esp. deep learning)
- Reflections on the consequences of technology



INDIGO Graffiti Project, [YouTube](#)

# Artificial Intelligence

- Already being used in projects, mainly: HTR/OCR (Transkribus, Escriptorium), NLP (spaCy)
- Many potential uses (many hitherto human tasks):
  - NLP (tagging, parsing; audio transcription; translation tasks)
  - Word Sense Disambiguation
  - AI for text encoding / annotating (editions)
  - Semantic Entity Linking
  - (Meta-)Data mining
  - Generative AI
  - Large Language Models (LLMs)
  - Vision Language Models (VLMs)
- Towards combining stochastic and knowledge-based approaches

## Training and keeping up-to-date

- Events
  - [ACDH-CH Events](#)
  - [ACDH-CH Event Series](#) (Tool Galleries, Research Days, ...)
  - University of Vienna DH Ringvorlesung ([2025S](#))
- Open Educational Resources (OER)
  - [DARIAH Campus](#)
  - [ACDH-CH How To](#)
- Internships ([at ACDH-CH](#))

## Mailing lists

- Humanist: [https://dhhumanist.org/membership\\_form.php](https://dhhumanist.org/membership_form.php)
- DHd-Mailingliste: <https://lists.uni-hamburg.de/mailman/listinfo/dhd>
- Linguist: <https://linguistlist.org/subscribe/>
- Corpora: <https://list.elra.info/mailman3/postorius/lists/corpora.list.elra.info/>
- DH Summer Institute: <https://lists.uvic.ca/mailman/listinfo/institute>
- TLDR (generally on tech): <https://tldr.tech>
- The Batch (on AI): <https://www.deeplearning.ai/the-batch/>

# Thank you! Questions?

<https://www.oeaw.ac.at/acdh>

<https://clariah.at>

<https://www.ditah.at>

<https://clarin.eu>

<https://dariah.eu>

<https://sshopencloud.eu>

Contact: [massimiliano.carloni](mailto:massimiliano.carloni@oeaw.ac.at) | [dimitra.grigoriou@oeaw.ac.at](mailto:dimitra.grigoriou@oeaw.ac.at)